



Review Test Submission: Assignment Week 7

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Course EBN 122 S2 2017

Test Assignment Week 7

Started 9/4/17 8:42 AM

Submitted 9/11/17 4:12 PM

Due Date 9/11/17 11:59 PM

Status Completed

Attempt Score 15 out of 20 points

Time Elapsed 175 hours, 30 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final answer correct to 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

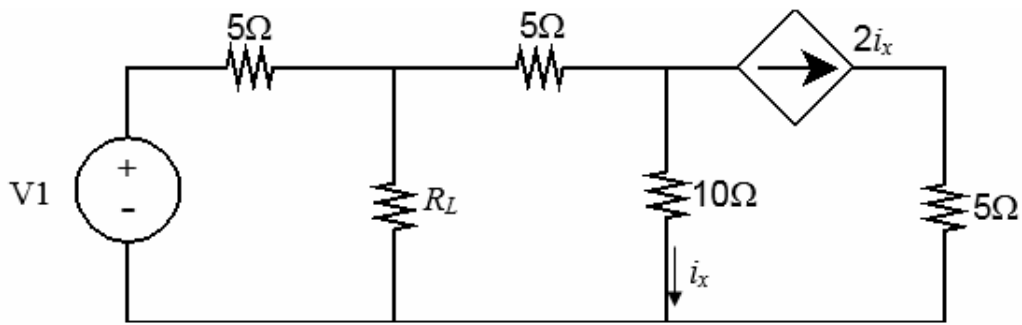
Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

0.5 out of 2 points



Determine the maximum power $P_{\max}$ (W), if $V_1 = 14$ V, $R_L = 3.125$ Ω .



Selected Answer: 1.53W

Correct Answer: $6.13 \pm 5\%$ (W)

Question 2

0.5 out of 2 points



The Thevenin equivalent of a circuit is $R_{TH} = 9\ \Omega$ and $V_{TH} = 15\text{ V}$. If a load resistance of $4\ \Omega$ is connected to the circuit, what is the power P_L absorbed by the load?

Selected Answer: 4.33W

Correct Answer: $5.33 \pm 5\%$ (W)

Question 3

2 out of 2 points



The Norton equivalent of a circuit is $I_N = 10\text{ mA}$ and

$R_N = 8\text{ k}\Omega$. What is the maximum power P_{max} that can be delivered to a load resistor?

Selected Answer: 200mW

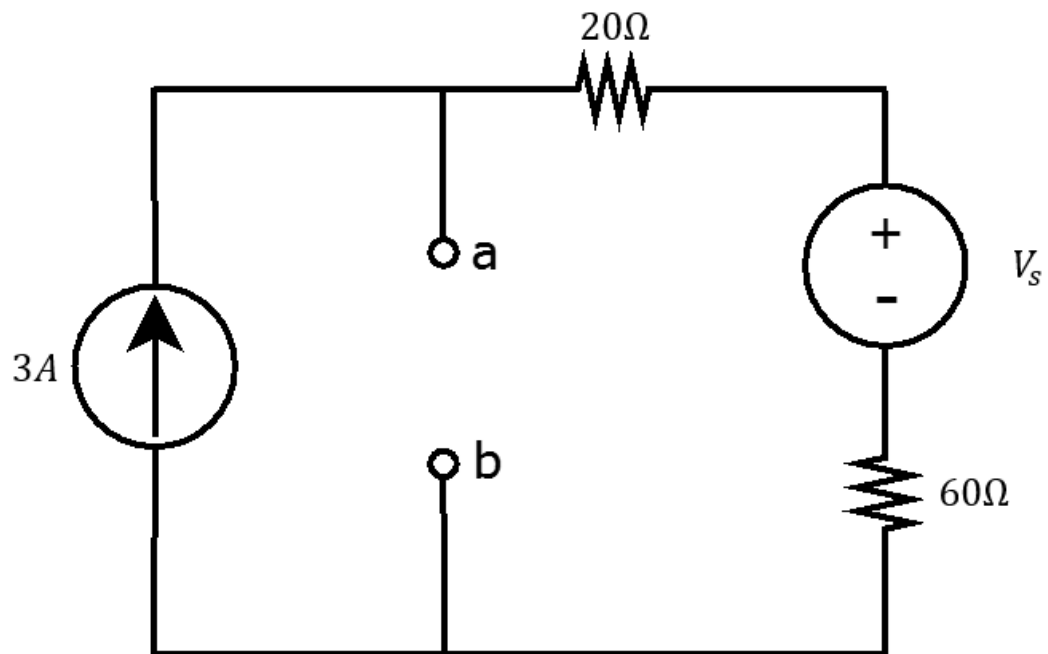
Correct Answer: $200.00 \pm 5\%$ (mW)

Question 4

2 out of 2 points



Determine the maximum power that can be delivered to an optimum resistor when it is connected between terminals a and b, given $V_s = 94$



Selected Answer: ☒ 349W

Correct Answer: ☒ $348.61 \pm 5\%$ (W)

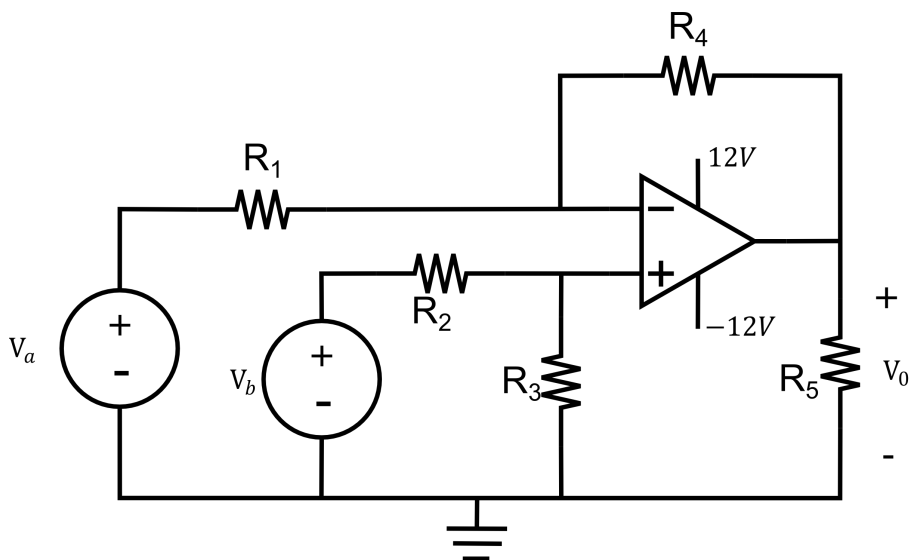
Question 5

2 out of 2 points



Find V_0 with

$R_1=28\Omega$, $R_2=6\Omega$, $R_3=29\Omega$, $R_4=62\Omega$, $R_5=29\Omega$, $V_a=6V$ and $V_b=2.5V$



Selected Answer: ☒ -6.63V

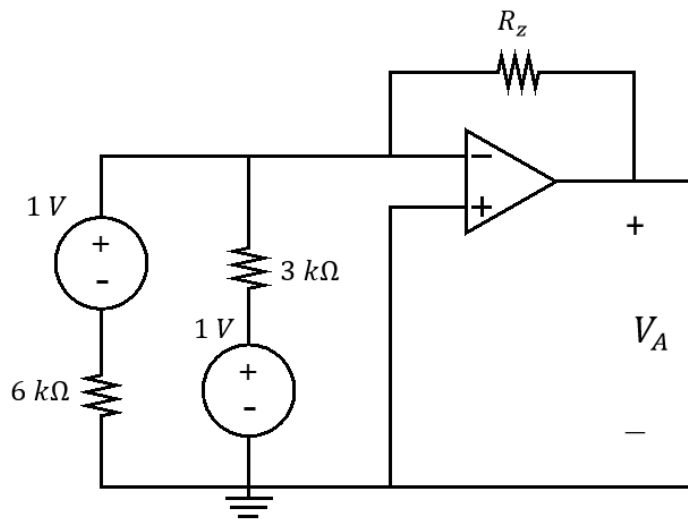
Correct Answer: ☒ $-6.628 \pm 5\%$ (V)

Question 6

2 out of 2 points



Given $R_Z = 15\text{ k}\Omega$, determine V_A .



Selected Answer: ☒ -7.5V

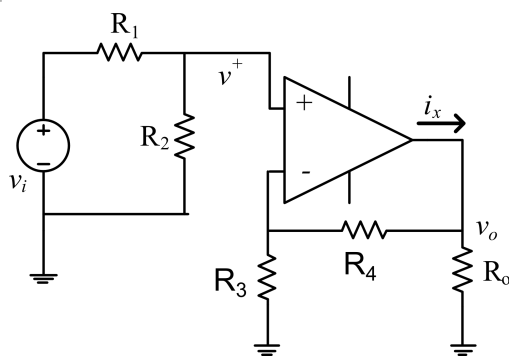
Correct Answer: ☒ -7.50 ± 5% (V)

Question 7

2 out of 2 points



Let $R_3 = 6 \text{ k}\Omega$ and $R_4 = 11 \text{ k}\Omega$. Find the current i_x when $R_0 = 807 \text{ }\Omega$ and $v^+ = -8 \text{ V}$.



Selected Answer: ☒ -29.4mA

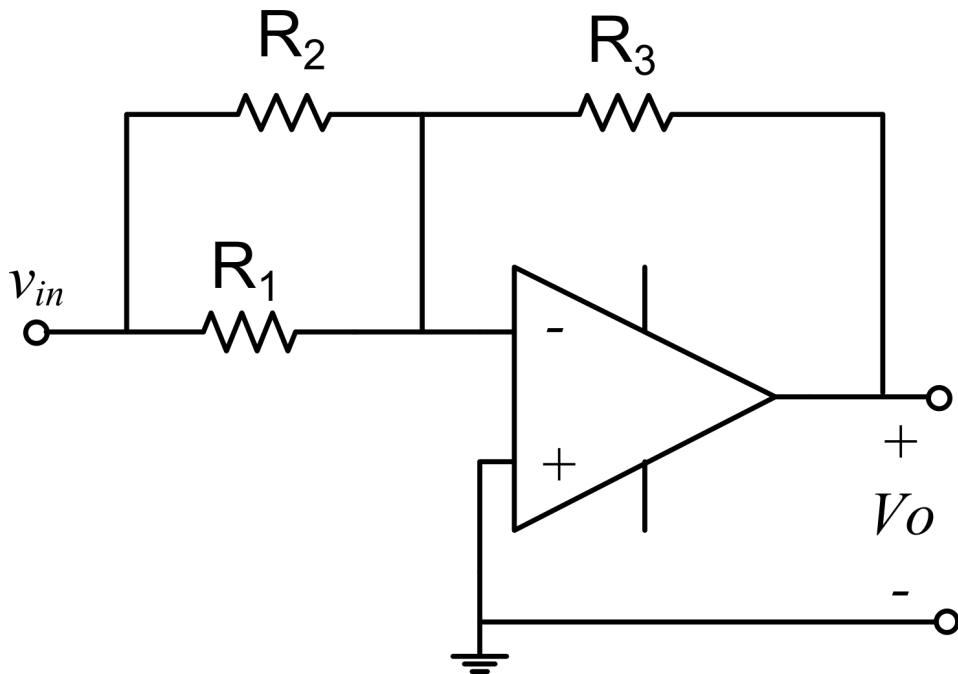
Correct Answer: ☒ -29.4209 ± 5% (mA)

Question 8

2 out of 2 points



Let $R_1 = 39 \text{ k}\Omega$, $R_2 = 15 \text{ k}\Omega$ and $R_3 = 93 \text{ k}\Omega$. Calculate the output voltage V_o , if $v_{in} = 13 \text{ mV}$



Selected Answer: ☒ -112mV

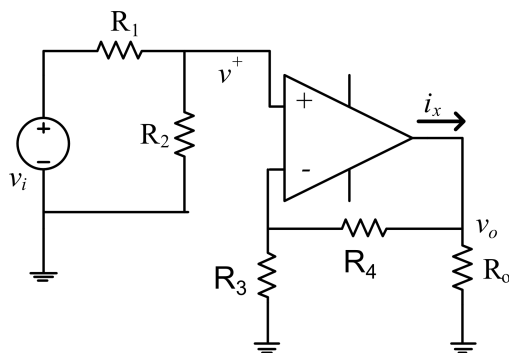
Correct Answer: ☒ -111.600 $\pm$ 5% (mV)

Question 9

0 out of 2 points



Let $R_3 = 9 \text{ k}\Omega$ and $R_4 = 56 \text{ k}\Omega$. Calculate the closed-loop gain (v_o/v_i) of the circuit when $R_1 = R_2$.



Selected Answer: ☒ 7.22

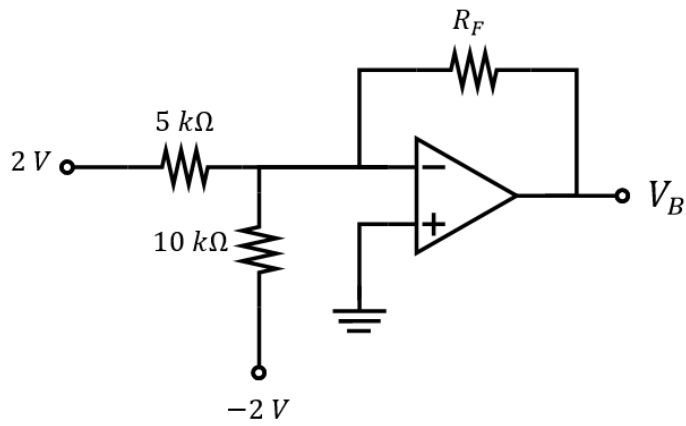
Correct Answer: ☒ 3.611 $\pm$ 5%

Question 10

2 out of 2 points



Determine R_F (k Ω) if $V_B = -5.1$,



Selected Answer: ☒ 25.5kohm

Correct Answer: ☒ $25.50 \pm 5\%$ (kohm)

Monday, September 11, 2017 4:13:02 PM SAST

← OK